

BERK KIVILCIM

PhD Student in Computer Science focusing on 3D Computer Vision and Geospatial Data

Portfolio | LinkedIn | berk.kivilcim@sabanciuniv.edu

EDUCATION

Sabanci University

Faculty of Engineering and Natural Sciences

2025 – Ongoing

Istanbul, Turkey

- **Fully Funded Ph.D. in Computer Science and Engineering, CGPA: 3.78**
- **Current Research Interest:** Hierarchical 3D Built Environment with Gaussian Splatting
- **Selected Courses:** Deep Learning (A), Machine Learning (A-)

Karlsruhe Institute of Technology (KIT)

Department of Civil Engineering, Geo and Environmental Sciences

2022 – 2024

Karlsruhe, Germany

- **Master of Science in Remote Sensing and Geoinformatics, GPA: 1.3** (German Grading Scale – *Very good*)
- **Thesis** [Computer Graphics and Visualization Group (TU Delft), Institute of Photogrammetry and Remote Sensing (KIT), 2024]: *Studying the Geometric Convergence Behaviour of Neural Radiance Fields (NeRFs) for Improving Training Time*
 - Proposed 3D geometric convergence and voxel based adaptive training termination strategy for NeRF.
 - Developed depth-guided pixel masking approach to focus training on unconverged regions.
 - Thesis, codes and implementation details shared on [GitHub](#)
- **Computer Vision Courses:** Deep Learning for Computer Vision (1.0), Computer Vision and Remote Sensing (1.3), Advanced Topics in Computer Vision (1.3), Active Sensors for Computer Vision (1.7),

Hacettepe University

Geomatics Engineering Department

2016 – 2021

Ankara, Turkey

- **Bachelor of Science in Geomatics Engineering, GPA: 3.01**
- **Graduation Project:** Investigation of Forest Fires with Google Earth Engine (as a group of three, 2021)
 - Reached the 2021 finals of TUBITAK 2242 project competition in the environment and energy field
- **Computer Vision Courses:** Data Structures and Algorithms (A1), Digital Image Processing (A2), Photogrammetric Image Analysis (A2), Photogrammetry (A3), Digital Imaging and Interpretation (B3), Fundamentals of Laser Scanning (B2), Signal Processing (C1)

PUBLICATIONS

- Kivilcim, B., Aptoula, E., Ozkan, H., **An Evolutionary Boost to Gaussian Splatting Optimization**, accepted by International Conference on Pattern Recognition (ICPR) BIOMAP workshop, **2026**
- Liu R., Kuper P.V., Kivilcim B., Breunig M: **Extending a Geodatabase Framework for AI-Based Analysis of Digital Elevation Models to Support Earth Observation Applications**, *International Conference on Data Science and Artificial Intelligence*. Singapore: Springer Nature Singapore, **2025**
https://doi.org/10.1007/978-981-95-4409-7_22
- Kivilcim, B., Bradley P.E: **A Data-Driven Approach for Urban Heat Island Predictions: Rethinking the Evaluation Metrics and Data Preprocessing**, *Urban Sci.* 2025, 9(5), 151,
<https://doi.org/10.3390/urbansci9050151>, **2025**
- Haitz, D., Kivilcim, B., Ulrich, M., Weinmann, M., and Weinmann, M.: **Density-based Geometric Convergence of NeRFs at Training Time: Insights from Spatio-temporal Discretization**, *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, XLVIII-2/W7-2024, 49–56,
<https://doi.org/10.5194/isprs-archives-XLVIII-2-W7-2024-49-2024>, **2024**

RESEARCH & WORK EXPERIENCE

Sabanci University

2025 - Ongoing

- **Teaching Assistant:** CS419 - Digital Image and Video Analysis (Fall 25/26), EE417 - Computer Vision (Spring 2026), CS300 Data Structures (Summer 2026)
- **Instructor:** Machine Learning and Artificial Intelligence (Summer 2026)

Karlsruhe Institute of Technology (KIT)

2022 – 2024

- **Research Assistant:** Worked on the Distributed Simulation of Processes in Buildings and City Models research project. Project Number: 469999674
- **Teaching Assistant:** “Numerical Mathematics” (Winter 23/24)

Akdeniz University

2020

- **Intern** at the Department of Space Sciences and Technologies: Researched various remote sensing and machine learning methodologies to analyze forest fires

Geotech Company

2019

- **Intern** at the Photogrammetry Department: Generated 3D surface and terrain models of the Medina

SKILLS & QUALIFICATIONS

- **Research Interests:** **Main Interests:** Deep Learning, Gaussian Splatting, Neural Radiance Fields, 3D Computer Vision, Geoinformatics
Other Expertise: Remote Sensing, Geodesy, Photogrammetry
- **Language:** Turkish (Native), English (Advanced)
IELTS Academic: Overall Score: **C1 - 7.0** (L: 7.5, R: 7.5, W: 6.0, S: 6.0)
YDS (English): 87.5
- **Activities:** Reviewer at IEEE Conference on Signal Processing and Communications Applications, 2026
President of capoeira student club at Hacettepe University, 2017 – 2019
Competitor in the Capoeira World Championship, senior category, 2018

REFERENCES

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| • Prof. Dr. Mustafa Türker | mturker@hacettepe.edu.tr | Tel: +90 312 297 6990 |
| • Assoc. Prof. Berk Anbaroğlu | banbar@hacettepe.edu.tr | Tel: +90 312 780 5791 |

MORE INFORMATION AVAILABLE ON **GitHub**
